



WINTRIX ACADEMY PRESENTS

DCET Previous Year Question Paper & Solutions

2025

DIPLOMA COMMON ENTRANCE TEST

GUIDANCE • STRATEGY • SUCCESS

100 Questions • Detailed Solutions • Subject-Wise Organisation

FIRST EDITION — 2026



DCET 2025 Previous Year Question Paper & Solutions

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About Wintrix Academy

Wintrix Academy is a premier DCET mentorship platform founded with a singular mission: to empower Diploma students across Karnataka with guidance, strategy, and resources to secure top ranks.



Mohammed Adnan

FOUNDER & CHIEF MENTOR

All-Karnataka DCET Rank 48 • 2nd Rank Hassan District • RVCE CSE (CGPA 9.19) • Placed at Red Hat

48

DCET RANK

9.8

DIPLOMA
CGPA

9.19

RVCE CGPA

50+

STUDENTS



Jnanesh Gowda

DIRECTOR & BRAND ARCHITECT

Turning confusion into clarity and effort into results through new-age learning strategies and results-engineered systems.

Preface

Dear DCET Aspirant,

This book contains the complete DCET 2025 Question Paper with detailed solutions for all 100 questions, organised subject-wise across five chapters.

The 2025 paper covers Engineering Mathematics, Statistics & Analytics, IT Skills, Electrical & Electronics Engineering, and Project Management Skills — each carrying 20 marks.

Attempt each question on your own first, then verify with the solution. Wishing you the very best!

Mohammed Adnan

Founder, Wintrix Academy

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01

CHAPTER ONE

Engineering Mathematics

20 Marks

Questions 1 – 20

Q.01

MATRICES

Choose the correct answer from the following:

- a) Every identity matrix is a scalar matrix
- b) Every scalar matrix is an identity matrix
- c) Every diagonal matrix is an identity matrix
- d) A square matrix whose each element is one is an identity matrix

✓ ANSWER: (A)

An identity matrix has 1's on the diagonal and 0's elsewhere. A scalar matrix has equal diagonal elements and 0's elsewhere. So every identity matrix is a scalar matrix (with scalar = 1).

Q.02

DETERMINANTS

The value of $\begin{vmatrix} 3 & 1 & -9 \\ -1 & 2 & 3 \\ 2 & 1 & -6 \end{vmatrix}$ is

- a) 45
- b) 90
- c) 0
- d) -45

✓ ANSWER: (C)

$$\begin{aligned} &= 3(-12-3) - 1(6-6) + (-9)(-1-4) \\ &= 3(-15) - 1(0) - 9(-5) \\ &= -45 - 0 + 45 = 0 \end{aligned}$$

Q.03

EIGENVALUES

The characteristic equation of $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ is

a) $\lambda^2 - 3\lambda - 2 = 0$

b) $\lambda^2 - 5\lambda + 10 = 0$

c) $\lambda^2 - 5\lambda - 2 = 0$

d) $\lambda^2 + 3\lambda + 10 = 0$

✓ ANSWER: (B)

$$|A - \lambda I| = 0$$

$$|(1-\lambda)(4-\lambda) - (-2)(3)| = 0$$

$$4 - \lambda - 4\lambda + \lambda^2 + 6 = 0$$

$$\lambda^2 - 5\lambda + 10 = 0$$

Q.04

MATRICES

If $|A| = \begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$, then $A \cdot \text{adj}(A)$ is equal to

a) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

b) $\begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$

✓ ANSWER: (D)

For any square matrix, $A \cdot \text{adj}(A) = |A| \cdot I$. Given $|A| = \begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$, so $A \cdot \text{adj}(A) = \begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$.

Q.05

STRAIGHT LINES

The slope of the straight line joining two points $(3, -4)$ and $(-1, 5)$ is

a) $\frac{4}{9}$

b) $-\frac{4}{9}$

c) $-\frac{1}{4}$

d) $-\frac{9}{4}$

✓ ANSWER: (D)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - (-4)}{-1 - 3} = \frac{9}{-4} = -\frac{9}{4}$$

The angle of inclination of the straight line is $\pi/4$ and y-intercept is 3, then equation of the straight line is

a) $y = x + 3$

b) $y = x - 3$

c) $y = -x - 3$

d) $y = -x + 3$

✓ ANSWER: (A)

$$\theta = \frac{\pi}{4} \rightarrow m = \tan\left(\frac{\pi}{4}\right) = 1, c = 3$$
$$y = mx + c = 1 \cdot x + 3 = x + 3$$

The equation of straight line passing through $(-5, 4)$ and parallel to X-axis is

a) $x - y + 9 = 0$

b) $x + 5 = 0$

c) $x + y + 1 = 0$

d) $y - 4 = 0$

✓ ANSWER: (D)

A line parallel to X-axis has slope $m = 0$. Passing through $(-5, 4)$: $y - 4 = 0$

What is the equation of the straight line passing through the points $(1, 5)$ and $(3, 2)$?

a) $7x - 4y + 13 = 0$

b) $3x + 2y - 17 = 0$

c) $3x + 2y - 13 = 0$

d) $3x + 2y + 13 = 0$

✓ ANSWER: (C)

$$m = (2-5)/(3-1) = -3/2$$
$$y - 5 = (-3/2)(x - 1)$$
$$2y - 10 = -3x + 3 \rightarrow 3x + 2y - 13 = 0$$

The ramp in front of a gate is elevated by an angle of $\frac{\pi}{3}$ radians, then the angle in degrees is

a) 15°

b) 75°

c) 60°

d) 30°

✓ ANSWER: (C)

$$\text{Degrees} = \text{Radians} \times \frac{180}{\pi} = \left(\frac{\pi}{3}\right) \times \left(\frac{180}{\pi}\right) = 60^\circ$$

If $\sin(90 - \theta) + \cos\theta = \sqrt{2} \cos(90 - \theta)$, then the value of $\operatorname{cosec}\theta$ is

a) $\frac{\sqrt{3}}{2}$

b) $\sqrt{\frac{3}{2}}$

c) $\sqrt{\frac{2}{3}}$

d) $\sqrt{3}$

✓ ANSWER: (B)

$$\cos\theta + \cos\theta = \sqrt{2} \sin\theta$$

$$2\cos\theta = \sqrt{2} \sin\theta \rightarrow \tan\theta = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\text{Opp} = \sqrt{2}, \text{Adj} = 1, \text{Hyp} = \sqrt{3}$$

$$\operatorname{cosec}\theta = \frac{\text{Hyp}}{\text{Opp}} = \frac{\sqrt{3}}{\sqrt{2}} = \sqrt{\frac{3}{2}}$$

If A, B, C are the angles of a triangle, then the incorrect relation is

a) $\sin\left(\frac{A+B}{2}\right) = \cos\left(\frac{C}{2}\right)$

b) $\cos\left(\frac{A+B}{2}\right) = \sin\left(\frac{C}{2}\right)$

c) $\tan\left(\frac{A+B}{2}\right) = \sin\left(\frac{C}{2}\right)$

d) $\cot\left(\frac{A+B}{2}\right) = \tan\left(\frac{C}{2}\right)$

✓ ANSWER: (C)

Since $A+B+C = 180^\circ$, $\frac{A+B}{2} = 90^\circ - \frac{C}{2}$. So $\tan\left(\frac{A+B}{2}\right)$ should equal $\cot\left(\frac{C}{2}\right)$, not $\sin\left(\frac{C}{2}\right)$. Option (c) is

incorrect.

The value of $\sin 40^\circ + \sin 20^\circ$ is

a) $\sqrt{3} \cos 10^\circ$

b) $\sqrt{3} \cos 20^\circ$

c) $\cos 10^\circ$

d) $\sqrt{2} \cos 10^\circ$

✓ ANSWER: (C)

$$\begin{aligned}\sin C + \sin D &= 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right) \\ &= 2\sin(30^\circ)\cos(10^\circ) \\ &= 2 \times \left(\frac{1}{2}\right) \times \cos 10^\circ = \cos 10^\circ\end{aligned}$$

Q.13

DIFFERENTIAL CALCULUS

The derivative of $y = \frac{1+x}{1-x}$ is

a) $\frac{2}{(1-x)^2}$

b) $\frac{-2}{(1-x)^2}$

c) $\frac{2}{1-x^2}$

d) $\frac{-2x}{(1-x)^2}$

✓ ANSWER: (A)

$$\begin{aligned} dy/dx &= \frac{(1-x)(1) - (1+x)(-1)}{(1-x)^2} \\ &= \frac{1-x+1+x}{(1-x)^2} = \frac{2}{(1-x)^2} \end{aligned}$$

Q.14

DIFFERENTIAL CALCULUS

If $y = \log(\sqrt{\tan x})$, then the value of dy/dx at $x = \frac{\pi}{4}$ is

a) ∞

b) 1

c) 0

d) $\frac{1}{2}$

✓ ANSWER: (B)

$$\begin{aligned} dy/dx &= \left(\frac{1}{\sqrt{\tan x}} \right) \times \left(\frac{1}{2\sqrt{\tan x}} \right) \times \sec^2 x \\ &= \frac{\sec^2 x}{2\tan x} \\ \text{At } x = \frac{\pi}{4}: \frac{(\sqrt{2})^2}{2 \cdot 1} &= \frac{2}{2} = 1 \end{aligned}$$

Q.15

DIFFERENTIAL CALCULUS

If $y = (2 + 3 \log x)^{10}$, then dy/dx at $x = 1$ is

a) $15(2^9)$

b) $(15 \times 2)^{10}$

c) $15(2^{10})$

d) $10(5^9)$

✓ ANSWER: (C)

$$\begin{aligned} dy/dx &= 10(2+3\log x)^9 \times \left(\frac{3}{x} \right) \\ \text{At } x=1 (\log 1=0): &= 10(2)^9 \times 3 = 30(2^9) \\ &= 15 \times 2 \times 2^9 = 15(2^{10}) \end{aligned}$$

Q.16

IMPLICIT DIFFERENTIATION

If $y = \sqrt{(\sin x + y)}$, then $dy/dx =$

a) $\frac{\cos x}{2y - 1}$

b) $\frac{\cos x}{1 - 2y}$

c) $\frac{\sin x}{2y - 1}$

d) $\frac{\sin x}{1 - 2y}$

✓ ANSWER: (A)

$$y^2 = \sin x + y \text{ (squaring both sides)}$$

$$2y(dy/dx) = \cos x + dy/dx$$

$$dy/dx(2y - 1) = \cos x$$

$$dy/dx = \frac{\cos x}{2y - 1}$$

Q.17

INTEGRAL CALCULUS

$\int \cot^2 x \, dx =$

a) $-\cot x - x + c$

b) $\cot x - x + c$

c) $\tan x - x + c$

d) $\tan^2 x + c$

✓ ANSWER: (A)

$$\cot^2 x = \operatorname{cosec}^2 x - 1$$

$$\int (\operatorname{cosec}^2 x - 1) dx = -\cot x - x + c$$

Q.18

DEFINITE INTEGRAL

The value of $\int_0^{\pi/4} \frac{\sec^2 x}{1 + \tan x} \, dx$ is

a) $\log 2 + 1$

b) $\log 1$

c) $\log 2$

d) $\log 2 - 1$

✓ ANSWER: (C)

$$\text{Let } f(x) = 1 + \tan x, \quad f'(x) = \sec^2 x$$

$$\int \frac{f'(x)}{f(x)} \, dx = \log |f(x)|$$

$$= [\log(1 + \tan x)]_0^{\pi/4}$$

$$= \log(1 + 1) - \log(1 + 0) = \log 2 - 0 = \log 2$$

$\int \sqrt{1 + \cos 2x} \, dx$ is

a) $2\cos x + c$

b) $2\sin x + c$

c) $\sqrt{2} \cos x + c$

d) $\sqrt{2} \sin x + c$

✓ ANSWER: (D)

$$1 + \cos 2x = 2\cos^2 x$$

$$\sqrt{2\cos^2 x} = \sqrt{2} \cos x$$

$$\int \sqrt{2} \cos x \, dx = \sqrt{2} \sin x + c$$

The value of $\int_0^{\pi/4} \sin x \cos x \, dx$ is

a) $-\frac{1}{4}$

b) $\frac{1}{4}$

c) $\frac{\pi}{2}$

d) $\frac{\pi}{4}$

✓ ANSWER: (B)

$$\int f(x) f'(x) \, dx = \frac{[f(x)]^2}{2}$$

$$f(x) = \sin x, f'(x) = \cos x$$

$$= \left[\frac{\sin^2 x}{2} \right]_0^{\pi/4} = \frac{\sin^2(\frac{\pi}{4})}{2} - 0$$

$$= \frac{(\frac{1}{\sqrt{2}})^2}{2} = \frac{\frac{1}{2}}{2} = \frac{1}{4}$$

02

CHAPTER TWO

Statistics & Analytics

20 Marks

Questions 21 – 40

Q.21

DATA

The data extracted from the Google forms has the file extension as

- a) .doc b) .docx
- c) .CSV d) .xlsx

✓ ANSWER: (C)

Data from Google Forms is exported as **.CSV** (Comma-Separated Values).

Q.22

DATA CLASSIFICATION

Match the following:

COLUMN A	COLUMN B
a) Data	i) Survey
b) Questionnaire	ii) Number of students
c) Interview	iii) Google form
d) Quantitative data	iv) Facts and figure

- a)** a-iv, b-i, c-iii, d-ii
- b)** a-iv, b-ii, c-iii, d-i
- c)** a-iv, b-iii, c-i, d-ii
- d)** a-iv, b-i, c-ii, d-iii

✓ ANSWER: (C)

Data = facts & figures (iv), Questionnaire = Google form (iii), Interview = method of survey (i), Quantitative = numerically measurable like number of students (ii).

Data cleaning refers to

- a) Modifying the gathered data
- b) Adding data to the gathered data
- c) Removing the data from gathered data
- d) Processing the data according to the defined problem statement

✓ ANSWER: (A/B/C/D)

All options are objectives of **data cleaning**: modifying, adding, removing, and processing data.

The syntax to find frequency distribution in an interval in Excel is

- a) =COUNTIFS(criteria_range1, criteria1,)
- b) =COUNTIF(criteria_range1, criteria1,)
- c) =COUNT(criteria_range1, criteria1,)
- d) =COUNTIFS(criteria1, criteria2,)

✓ ANSWER: (A)

The syntax is **=COUNTIFS(criteria_range1, criteria1,)**

Which tab in Excel supports in plotting Pie chart?

- a) Home tab
- b) View tab
- c) Insert tab
- d) Data tab

✓ ANSWER: (C)

The **Insert tab** is used to plot pie charts in Excel.

Q.26**CHARTS**

Frequency polygon is a _____ graph

- a) Scatter graph
- b) Pie graph
- c) Bar graph
- d) Line graph

✓ ANSWER: (D)

A frequency polygon is a **line graph**.

Q.27**HISTOGRAM**

Which of the following statements on Histogram is true?

- a) A Histogram can display categorical data effectively
- b) The bars in a Histogram should not touch each other
- c) A Histogram shows the frequency distribution of continuous data
- d) A Histogram is same as a bar chart

✓ ANSWER: (C)

A histogram shows **frequency distribution of continuous data**.

Choose the right stem-leaf plot for data: 52, 20, 44, 28, 20, 41, 30, 56, 32, 28

PLOT-1:

STEM	LEAF
2	0088
3	20
4	14
5	26

PLOT-2:

STEM	LEAF
2	0088
3	02
4	14
5	26

PLOT-3:

STEM	LEAF
2	08
3	20
4	14
5	26

PLOT-4:

STEM	LEAF
2	80
3	20
4	41
5	62

a) Plot-1

b) Plot-2

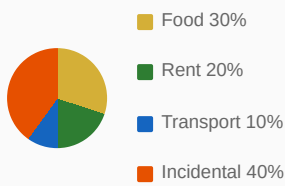
c) Plot-3

d) Plot-4

✓ ANSWER: (B)

Arranging data: 20,20,28,28,30,32,41,44,52,56. Stem|Leaf: 2|0088, 3|02, 4|14, 5|26. Key: 2|0=20.

From a pie chart: Food 30%, Rent 20%, Transport 10%, Incidental 40%. Salary = ₹50,000. Total expenditure on rent and transport is?



- a) ₹30,000 b) ₹25,000
- c) ₹15,000 d) ₹10,000

✓ ANSWER: (C)

$$\begin{aligned} \text{Rent} + \text{Transport} &= 20\% + 10\% = 30\% \\ &= \frac{30}{100} \times 50000 = ₹15,000 \end{aligned}$$

For data: 28, 36, 44, 58, 44, 33, 76, 59, 58, 39, 58. Match the following:

MEASURE	VALUE
i) Range	a) 58
ii) Mean	b) 44
iii) Mode	c) 48.4545
iv) Median	d) 48

- a) i-d, ii-c, iii-b, iv-a b) i-d, ii-c, iii-a, iv-b
- c) i-a, ii-b, iii-c, iv-d d) i-d, ii-a, iii-b, iv-c

✓ ANSWER: (B)

$$\begin{aligned} \text{Range} &= 76 - 28 = 48 \text{ (d)} \\ \text{Mean} &= \frac{533}{11} = 48.4545 \text{ (c)} \\ \text{Mode} &= 58 \text{ (most frequent) (a)} \\ \text{Median} &= 44 \text{ (middle value) (b)} \end{aligned}$$

Q.31

EXCEL

The Excel syntax for finding quartile 4 (4th quartile) for a given data set is

- a) =QUARTILE [array, 4] b) =QUARTILE (array, 4)
- c) =QUARTILE (array_4) d) =QUARTILE (array-4)

✓ ANSWER: (B)

The syntax is =QUARTILE(array, 4)

Q.32

STATISTICS

Which of the following equalities and inequalities is true for the data set 25, 25, 25, 25, 25, 25, 25, 25?

- a) Variance > Standard deviation b) Variance = Standard deviation
- c) Variance < Standard deviation d) Variance < 0, Standard deviation < 0

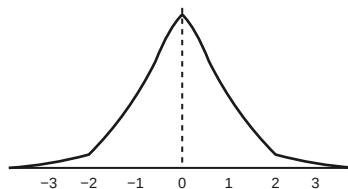
✓ ANSWER: (B)

All values are identical, so Variance = SD = 0. Hence Variance = Standard deviation.

Q.33

DISTRIBUTION

Skewness of the symmetric distribution shown is



- a) > 0 b) = 0
- c) < 0 d) = 1

✓ ANSWER: (B)

In a symmetric distribution, skewness $\beta_1 = 0$.

Q.34

KURTOSIS

If the Kurtosis = 3, then the data distribution is called

- a) Leptokurtic **b) Mesokurtic**
- c) Platykurtic d) Skewkurtic

✓ ANSWER: (B)

When $\beta_2 = 3$, the distribution is **mesokurtic** (normal distribution).

Q.35

PYTHON

Assertion (A): An "int" type data in Python can store floating point numbers.

Reasoning (R): Python "int" type only stores positive integers.

- a) A is true, R is false b) A is false, R is true
- c) A is false, R is false** d) A is true, R is true

✓ ANSWER: (C)

A is false — int cannot store floats. R is also false — int stores positive, negative, and zero.

Q.36

PYTHON

Match the data types:

EXPRESSION	DATA TYPE
a) a = 5	i) String
b) a or b	ii) Float
c) "Hello KEA"	iii) Integer
d) C = 5.9	iv) Boolean

- a) a-iii, b-iv, c-i, d-ii** b) a-iv, b-i, c-ii, d-iii
- c) a-iv, b-i, c-iii, d-ii d) a-i, b-ii, c-iii, d-iv

✓ ANSWER: (A)

a=5 → Integer(iii), a or b → Boolean(iv), "Hello KEA" → String(i), C=5.9 → Float(ii)

Q.37

PYTHON

What is the syntax to assign the data type to a variable in Python?

- a) Name = "hello world" b) Height = 5.9
- c) Assertion = True **d) All of the above**

✓ ANSWER: (D)

Python uses **dynamic typing** — data types are assigned automatically based on the value.

Q.38

PYTHON

Which of the following is a valid variable name in Python?

- a) 1 Var **b) Var_name**
- c) Var - name d) Var name

✓ ANSWER: (B)

Var_name is valid — starts with letter, contains only letters and underscore, no spaces or special characters.

Q.39

PYTHON

Output of the following Python program is:

```
n = 5
A = 0
while (n > 0) :
    A = A + n
    n = n - 1
print (A)
```

- a) 15** b) 5
- c) 55 d) 0

✓ ANSWER: (A)

```
n=5: A=0+5=5, n=4: A=5+4=9,
n=3: A=9+3=12, n=2: A=12+2=14,
n=1: A=14+1=15, n=0: loop stops.
print(A) → 15
```

Which of the following is the correct syntax of an if statement in Python?

a) if x>0 then:

b) if (x>10)

c) if x>10:

d) if x>10;

✓ ANSWER: (C)

Python if syntax: `if condition:` — no parentheses needed, must end with colon.

03

CHAPTER THREE

IT Skills

20 Marks

Questions 41 – 60

Q.41

ALGORITHMS

Consider: Step 1: $A=5$, $B=3$. Step 2: if $A>B$, then $A=A-B$. Step 3: $B=A+B$. Step 4: Output B . What is the output?

a) 2

b) 5

c) 8

d) 3

✓ ANSWER: (B)

$A=5$, $B=3$. $A>B$? Yes $\rightarrow A=5-3=2$

$B = A+B = 2+3 = 5$. Output $B = 5$

Q.42

ALGORITHMS

Which of the following best defines an algorithm?

a) A graphical representation

b) A code snippet

c) A step-by-step procedure to solve a problem

d) A mobile application

✓ ANSWER: (C)

An algorithm is a **finite sequence of well-defined steps** to solve a problem.

Consider the following algorithm:

Start

if $a \leq b$ and $a \leq c$

 print a

else if $b \leq a$ and $b \leq c$

 print b

else

 print c

End.

if $a = 10$, $b = 25$, $c = 33$

What output will be printed for the above algorithm?

a) 25

b) 10

c) 33

d) None of the above

✓ ANSWER: (B)

$10 \leq 25$ (True) and $10 \leq 33$ (True) → both conditions true → prints 10.

Consider the following algorithm:

if $\text{num} > 0$

$\text{num} = \text{num} + 10$

 print num

else if $\text{num} == 0$

$\text{num} = \text{num} * 10$

 print num

else

 print num.

What will be the output, if $\text{num} = -5$?

a) 50

b) -50

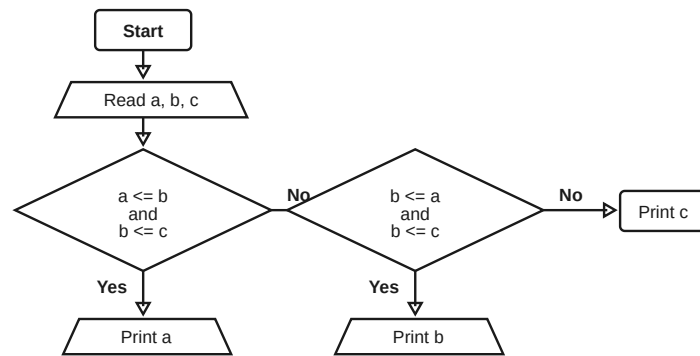
c) -5

d) 5

✓ ANSWER: (C)

$-5 > 0$? No. $-5 == 0$? No. else: print num → prints -5.

Consider the following flowchart.



What will be the output if $a = 10$, $b = 15$, $c = 5$?

- a) 10 b) 15
c) error d) 5

✓ ANSWER: (D)

$a \leq b$ and $b \leq c \rightarrow 10 \leq 15$ (T) and $15 \leq 5$ (F) $\rightarrow$ False. $b \leq a$ and $b \leq c \rightarrow 15 \leq 10$ (F) $\rightarrow$ False. else $\rightarrow$ print $c = 5$.

Which of the following statements about web browsers is correct?

- a) Web browsers translate HTML, CSS and JavaScript into machine code before displaying a web page to the user.
- b) Web browsers can only display content written in HTML and CSS, but they cannot execute JavaScript.
- c) Web browsers interpret HTML, CSS & JavaScript to render a webpage allowing dynamic content and interaction.
- d) Web browsers use server-side programming languages like PHP to render webpages directly

✓ ANSWER: (C)

Browsers **interpret HTML, CSS & JavaScript** to render webpages.

Consider the following HTML code snippet.

```
<select name = "cars" id = "carselect">
  <option value = "volvo">Volvo</option>
  <option value = "XUV">XUV</option>
  <option value = "Tata">Tata</option>
</select>
```

What does the <select> element do in this code?

- a) It creates a text input field for users to type in
- b) It creates a dropdown list from which the user can select a car brand.**
- c) It allows users to choose multiple options from a list of cars.
- d) It creates a button that users can click to select a car brand

✓ ANSWER: (B)

The <select> element creates a dropdown list.

Which CSS property is used to set the background color of a webpage?

- a) background-color**
- b) color
- c) bg-color
- d) background-image

✓ ANSWER: (A)

background-color sets the background color of an element.

Which CSS selector is used to select an HTML element with a specific id?

- a) .id
- b) #id**
- c) id
- d) [id]

✓ ANSWER: (B)

The hash symbol # is the ID selector in CSS.

Which event is triggered when a user clicks on an HTML element in JavaScript?

- a) onload
- b) onclick
- c) onhover
- d) onchange

✓ ANSWER: (B)

The **onclick** event fires when a user clicks on an element.

Which factor can help improve a website's SEO ranking?

- a) Using JavaScript for content rendering
- b) High Keyword density in the content
- c) Using relevant and high-quality backlinks
- d) Including as many images as possible

✓ ANSWER: (C)

Backlinks from reputable websites are "votes of confidence" for SEO.

Which of the following are the correct description with respect to structured organisation.

- (a) Operations → produces goods or services, ensures quality and efficiency.
- (b) Marketing → Identifies customer needs, promotes or services.
- (c) H.R. (Human Resources): Manages budgeting, accounting and financial planning.

- a) (a), (b)
- b) (a), (b), (c)
- c) (a) only
- d) (b) only

✓ ANSWER: (A)

HR does not handle budgeting/financial planning — that's Finance. Only **(a) Operations and (b) Marketing** are correct.

Which of the following statements best describes the purpose of ERP tools in an organisation?

- a) ERP tools are designed to manage customer relationships and marketing strategies.
- b) ERP tools are used to integrate and automate key business processes across department as finance, HR, inventory and supply chain.**
- c) ERP tools are primarily focused in managing email communication between employees an organization.
- d) ERP tools are closely used for managing the IT infrastructure and hardware resources organisation.

✓ ANSWER: (B)

ERP systems **unify operations across departments** for seamless coordination.

In cloud computing, data is stored in

- a) Local folders
- b) Physical notebooks
- c) Remote servers accessed via the internet**
- d) USB drives

✓ ANSWER: (C)

Cloud computing stores data on **remote internet-connected servers**.

Which of the following is an example of IoT in smart homes?

- a) A GPS system in a car for navigation
- b) A thermostat that adjusts temperature based on occupancy and preferences**
- c) An e-commerce website for shopping
- d) A manual doorbell

✓ ANSWER: (B)

Smart **thermostats** use sensors and connectivity — a core IoT feature in smart homes.

Which of the following is an example of Software-as-a-Service (SaaS)?

- a) Microsoft Azure
- b) Google Drive
- c) Amazon Web Services (AWS) EC2
- d) Google Kubernetes Engine

✓ ANSWER: (A & B)

SaaS delivers applications over the internet. **Microsoft Azure** also provides SaaS, and **Google Drive** is a SaaS application.

Which of the following is a platform for developing IoT applications?

- a) Google Docs
- b) Arduino
- c) Microsoft
- d) WinZip

✓ ANSWER: (B)

Arduino is an open-source electronics platform for building IoT devices.

Ravi often uses incognito mode while browsing, believing it protects him from all types of tracking and threats. He does not use antivirus software or phishing protection.

Which of the following statement about Ravi's assumption is correct?

- a) Incognito mode provides full malware protection
- b) Incognito mode hides activity from the device and protects against all trackers
- c) Incognito mode only prevents local history tracking not online threats
- d) Incognito mode encrypts his traffic and login credentials

✓ ANSWER: (C)

Incognito mode only stops saving local history, cookies, and site data. It does **not protect** from malware or phishing.

Rita uses a public computer at a cyber cafe for email and closes the browser without logging out. What cyber safety mistake did she make?

- a) Read personal emails in public
- b) Used a public computer
- c) She didn't logout of her account**
- d) She didn't clear browsing history

✓ ANSWER: (C)

Not logging out on public computers poses a risk of unauthorized access.

To safeguard against cyber attacks that steal sensitive information, which browser setting should be enabled?

- a) Enable automatic updates
- b) Turn on phishing and malware protection**
- c) Disable phishing and malware protection
- d) Enable cookies

✓ ANSWER: (B)

Phishing and malware protection warns users about harmful websites and downloads.

04

CHAPTER FOUR

Fundamentals of Electrical & Electronics Engineering

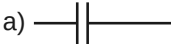
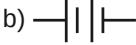
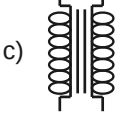
20 Marks

Questions 61 – 80

Q.61

SYMBOLS

Match the following with symbols

COMPONENT	SYMBOL
i) Resistance	a) 
ii) Transformer	b) 
iii) Capacitor	c) 
iv) Battery	d) 

a) i-a, ii-c, iii-d, iv-b

b) i-d, ii-b, iii-c, iv-a

c) i-b, ii-c, iii-d, iv-a

d) i-d, ii-c, iii-a, iv-b

✓ ANSWER: (D)

Resistance → zigzag, Transformer → coils, Capacitor → parallel lines, Battery → long/short lines.

Q.62

EARTHING

Earthing plate is made up of

a) Bakelite

b) Aluminum

c) Copper

d) Plastic

✓ ANSWER: (C)

Copper is a good conductor and corrosion-resistant, ideal for earthing.

Match the following:

QUANTITY	UNIT
i) Current	a) Watts
ii) Power	b) Hertz
iii) Energy	c) kWh
iv) Frequency	d) Ampere

a) i-a, ii-b, iii-d, iv-c

b) i-a, ii-c, iii-b, iv-d

c) i-c, ii-b, iii-d, iv-a

d) i-d, ii-a, iii-c, iv-b

✓ ANSWER: (D)

Current → Ampere, Power → Watts, Energy → kWh, Frequency → Hertz.

Q.64

CIRCUITS

If 12Ω and 6Ω are connected in parallel, the equivalent resistance is

a) 18Ω

b) 06Ω

c) 04Ω

d) 02Ω

✓ ANSWER: (C)

$$\frac{1}{R} = \frac{1}{12} + \frac{1}{6} = \frac{1+2}{12} = \frac{3}{12} = \frac{1}{4}$$

$$R = 4\Omega$$

Q.65

OHM'S LAW

In a circuit with $R=6\Omega$ and $V=12V$, the current is

a) $72A$

b) $0.5A$

c) $2A$

d) $1.2A$

✓ ANSWER: (C)

$$I = \frac{V}{R} = \frac{12}{6} = 2A$$

AC THEORY

a) 100 Hz b) 200 Hz

c) 50 Hz d) 150 Hz

$$f = \frac{1}{T} = \frac{1}{0.02} = 50 \text{ Hz}$$

AC THEORY

a) Both current and voltage are in phase

b) Current leads the voltage

c) Voltage lags the current

d) Both are out of phase

In a pure resistive AC circuit, current and voltage are **in phase**.

PROTECTION

a) 250 mA b) 500 mA
c) 1 A d) 1250 A

HRC (High Rupturing Capacity) fuses handle very high fault currents, up to **1250 A or more**.

WIRING

a) Double pole

b) Two way

c) Single pole

d) One way

Two-way switches allow control from two different locations.

Q.70**FUSE**

A fuse is always connected _____ to the circuit.

- a) Parallel
- b) Across the load
- c) Series**
- d) Across the supply

✓ ANSWER: (C)

A fuse is connected in **series** to break the circuit on overload.

Q.71**TRANSFORMER**

In a Transformer, _____ remains constant

- a) Voltage
- b) Frequency**
- c) Current
- d) Both current and voltage

✓ ANSWER: (B)

In both step-up and step-down transformers, **frequency** remains the same (e.g., 50 Hz).

Q.72**BATTERIES**

The combination of two or more cells is

- a) Battery**
- b) load cell
- c) Primary cell
- d) Generator

✓ ANSWER: (A)

A **battery** is formed by connecting two or more cells.

Match the following:

	COLUMN A		COLUMN B
i)	Ampere hour	a)	1 ϕ Induction motor
ii)	KVA	b)	Lead acid battery
iii)	Star delta	c)	Transformer
iv)	Fan	d)	3 ϕ Induction motor

a) i-a, ii-b, iii-c, iv-d

b) i-c, ii-a, iii-d, iv-b

c) i-b, ii-c, iii-d, iv-a

d) i-b, ii-a, iii-c, iv-d

✓ ANSWER: (C)

Ampere-hour → Battery, KVA → Transformer, Star-delta → 3 ϕ motor, Fan → 1 ϕ motor.

Q.74

TRANSFORMER

In a transformer, the winding to which the load is connected is called _____ winding

a) Primary

b) Axillary

c) Secondary

d) Concentric

✓ ANSWER: (C)

The **secondary winding** is connected to the load and delivers output voltage.

Q.75

TRANSFORMER

Which statement is FALSE with respect to transformer?

a) Converts high voltage to low voltage

b) Converts high frequency to low frequency

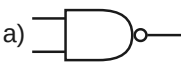
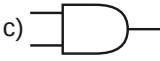
c) Converts low voltage to high voltage

d) Transfers energy at constant frequency

✓ ANSWER: (B)

A transformer does **not change frequency**. It can step up or step down voltage, but frequency remains unchanged.

Match the following logic gates with their symbols:

GATE	SYMBOL
i) AND	a) 
ii) OR	b) 
iii) NAND	c) 
iv) NOR	d) 

a) i-a, ii-b, iii-c, iv-d

b) i-b, ii-c, iii-d, iv-a

c) i-d, ii-a, iii-c, iv-b

d) i-c, ii-d, iii-a, iv-b

✓ ANSWER: (D)

AND, OR, NAND, NOR matched with their respective standard symbols.

Which is a common type of sensor used for measuring temperature?

a) Proximity sensor

b) Pressure sensor

c) Flow sensor

d) Thermocouple

✓ ANSWER: (D)

A **thermocouple** converts thermal energy into electrical voltage for temperature measurement.

What does the fourth band in a 4-band resistor indicate?

a) Resistance

b) Multiplier

c) Power rating

d) Tolerance

✓ ANSWER: (D)

1st & 2nd bands = significant digits, 3rd = multiplier, 4th = **tolerance**.

Which of the following is an example of Insulator?

- a) Aluminium
- b) Rubber
- c) Iron
- d) Silicon

✓ ANSWER: (B)

Rubber resists the flow of electric current. Aluminium and Iron are conductors; Silicon is a semiconductor.

Diode is made up of _____ material.

- a) Conducting
- b) Semiconducting
- c) Insulating
- d) Electronic

✓ ANSWER: (B)

Diodes are made from **semiconductor** materials like Silicon or Germanium.

05

CHAPTER FIVE

Project Management Skills

20 Marks

Questions 81 – 100

Q.81

ANALOGY

Complete the analogy: Project : Deliverables :: Operation : _____

- a) Deadlines
- b) Efficiency
- c) Repetition**
- d) Conclusion

✓ ANSWER: (C)

Projects have specific deliverables; operations are permanent processes with **repetitive** outcomes.

Q.82

OPERATIONS

Multiple objectives are achieved again and again in _____

- a) Operation**
- b) Project
- c) Business
- d) Sales

✓ ANSWER: (A)

Operations are repetitive processes where objectives are achieved again and again.

Q.83

CONSULTANCY

Which are examples of consultancy firms? A) Accenture, B) TCS, C) Tata Motors, D) Birla Technical Services

- a) A, B and C only
- b) A, B and D only**
- c) A and D only
- d) A, B, C and D

✓ ANSWER: (B)

Accenture, TCS, and Birla Technical Services are consultancy firms. **Tata Motors** is automotive manufacturing.

Which of the following is NOT an obstacle in project management?

- a) Undefined project scope
- b) Poor communication among team members
- c) Clear project objectives and goals**
- d) Limited resources

✓ ANSWER: (C)

Clear project objectives is an advantage, not an obstacle.

Project Procedure Manual : Standard Operating Procedures :: Project Diary : _____

- a) Records daily events and issues**
- b) Provides step-by-step instructions
- c) Outlines rules and processes
- d) Used only during project closure

✓ ANSWER: (A)

A project diary **records daily events, activities, and issues** throughout the project life cycle.

Match the following:

TEAM TYPE	ROLE
i) Initial project team	a) Guides the project with advice and support
ii) Core project team	b) Works on the project every day, making decisions and doing tasks
iii) Project stakeholders	c) People interested in the project outcomes like clients or sponsors
iv) Project advisors	d) Helps to start the project and plan the initial steps

- a) i-d, ii-b, iii-c, iv-a**
- b) i-a, ii-b, iii-c, iv-d
- c) i-b, ii-a, iii-c, iv-d
- d) i-d, ii-c, iii-b, iv-a

✓ ANSWER: (A)

Initial team starts the project (d), Core team works daily (b), Stakeholders are interested parties (c), Advisors guide with advice (a).

What does Work Breakdown Structure (WBS) do to help a Project Manager?

- a) Track the project's financial performance
- b) Breakdown the project into smaller, manageable tasks**
- c) Determine project's risk level
- d) Allocate resources to team members

✓ ANSWER: (B)

WBS **breaks down complex work into smaller, manageable tasks**.

Which is NOT a subpart of Project Execution Plan (PEP)?

- a) Organizational plan
- b) Work packing plan
- c) Staffing plan**
- d) Contracting plan

✓ ANSWER: (C)

Staffing plan is not included in PEP.

High demand, high selling price, low variable cost are examples of

- a) Normal Scenario
- b) Worst Scenario
- c) Best Scenario**
- d) Abnormal Scenario

✓ ANSWER: (C)

High demand, high selling price, low cost represent the **best scenario**.

In _____ phase, project definition and problem identification are defined

- a) Planning
- b) Execution
- c) Initiation**
- d) Closure

✓ ANSWER: (C)

Initiation is the first phase where project is defined and problems are identified.

Q.91**RISK**

Risk arising due to fluctuation of interest rate OR mistakes in accounting is known as _____ risks.

- a) Policy
- b) Technical
- c) Marketing
- d) Financial

✓ ANSWER: (D)

Financial risks arise from interest rate fluctuations or accounting errors.

Q.92**ASSERTION**

Assertion (A): Time overrun can lead to cost overrun in a project.

Reason (R): More time often means more money spent on labour and resources.

- a) Both (A) and (R) are true and (R) is the correct explanation of (A)
- b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- c) (A) is true, but (R) is false
- d) (A) is false, but (R) is true

✓ ANSWER: (A)

Both statements are true, and the reason correctly explains why time overrun leads to cost overrun.

Q.93**CPM**

The full form of CPM in project management is

- a) Critical Process Management
- b) Critical Path Method
- c) Continuous Project Monitoring
- d) Cost Planning Model

✓ ANSWER: (B)

CPM stands for **Critical Path Method**.

Q.94

TOOLS

Which is NOT a tool of project planning, scheduling and monitoring?

- a) Gantt Chart
- b) Flow chart
- c) PERT Chart
- d) Milestone Chart

✓ ANSWER: (B)

Flow chart is not a project planning/scheduling/monitoring tool.

Q.95

EVALUATION

Project Evaluation can be done _____

- a) At the end of project only
- b) Throughout the project lifecycle
- c) Only during planning phase
- d) At the start only

✓ ANSWER: (B)

Project evaluation is done **throughout the project lifecycle**.

Q.96

PLANNING

Identify the incorrect statement regarding project planning functions:

- a) Defines project goals and objectives
- b) Allocates resources without considering constraints
- c) Establishes a project timeline and milestones
- d) Identifies potential risks and mitigates them

✓ ANSWER: (B)

Project planning allocates resources **by considering constraints**, not without.

Q.97

SMART

Which is NOT a key component of "SMART" goal?

- a) Time-Bound
- b) Random
- c) Achievable
- d) Specific

✓ ANSWER: (B)

SMART = Specific, Measurable, Achievable, Realistic, Timely. **Random** is not a component.

The process of correcting and rectifying the errors in the project is

- a) Project Audit
- b) Project Monitoring
- c) Project Evaluation
- d) Project Review

✓ ANSWER: (D)

Project Review is the process of correcting and rectifying errors.

Optimistic Time = 4 days, Pessimistic Time = 8 days, Most likely time = 6 days. The expected time to complete is

- a) 7 days
- b) 6 days
- c) 5 days
- d) 4 days

✓ ANSWER: (B)

$$T_e = \frac{t_o + 4t_m + t_p}{6}$$
$$= \frac{4 + 4(6) + 8}{6} = \frac{36}{6} = 6 \text{ days}$$

Watching 3D movie is an example of

- a) Augmented Reality
- b) Virtual Reality
- c) Augmented and Virtual Reality
- d) Cloud Technology

✓ ANSWER: (A, B, C)

Watching a 3D movie is an example of **Augmented Reality**, **Virtual Reality**, and both combined.



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